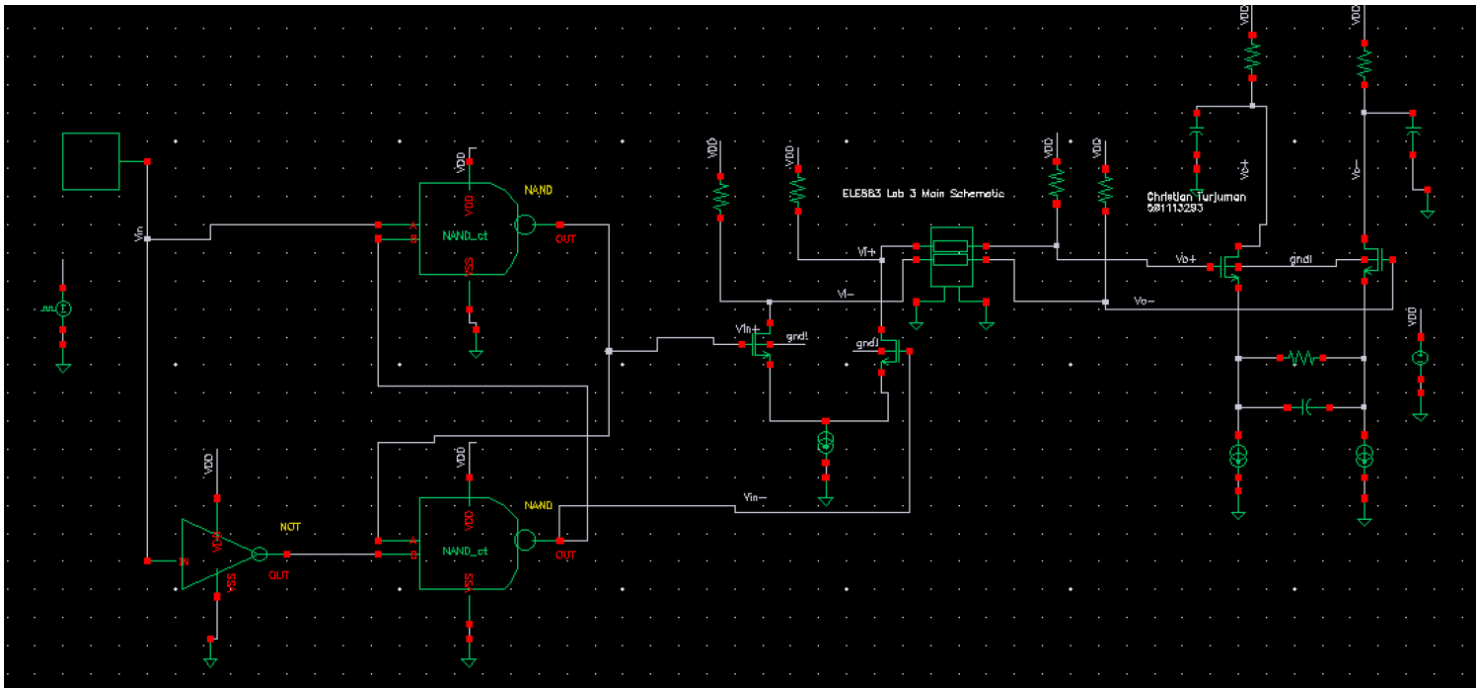
Main Schematic:



Transistor Sizing:

Transistor	Size
Taps	3.3u
CTLE transistors	16u

Load Resistor (RL) Calculation:

Sizing top 2 resistors

1Ghz= 1 / 2pi RC

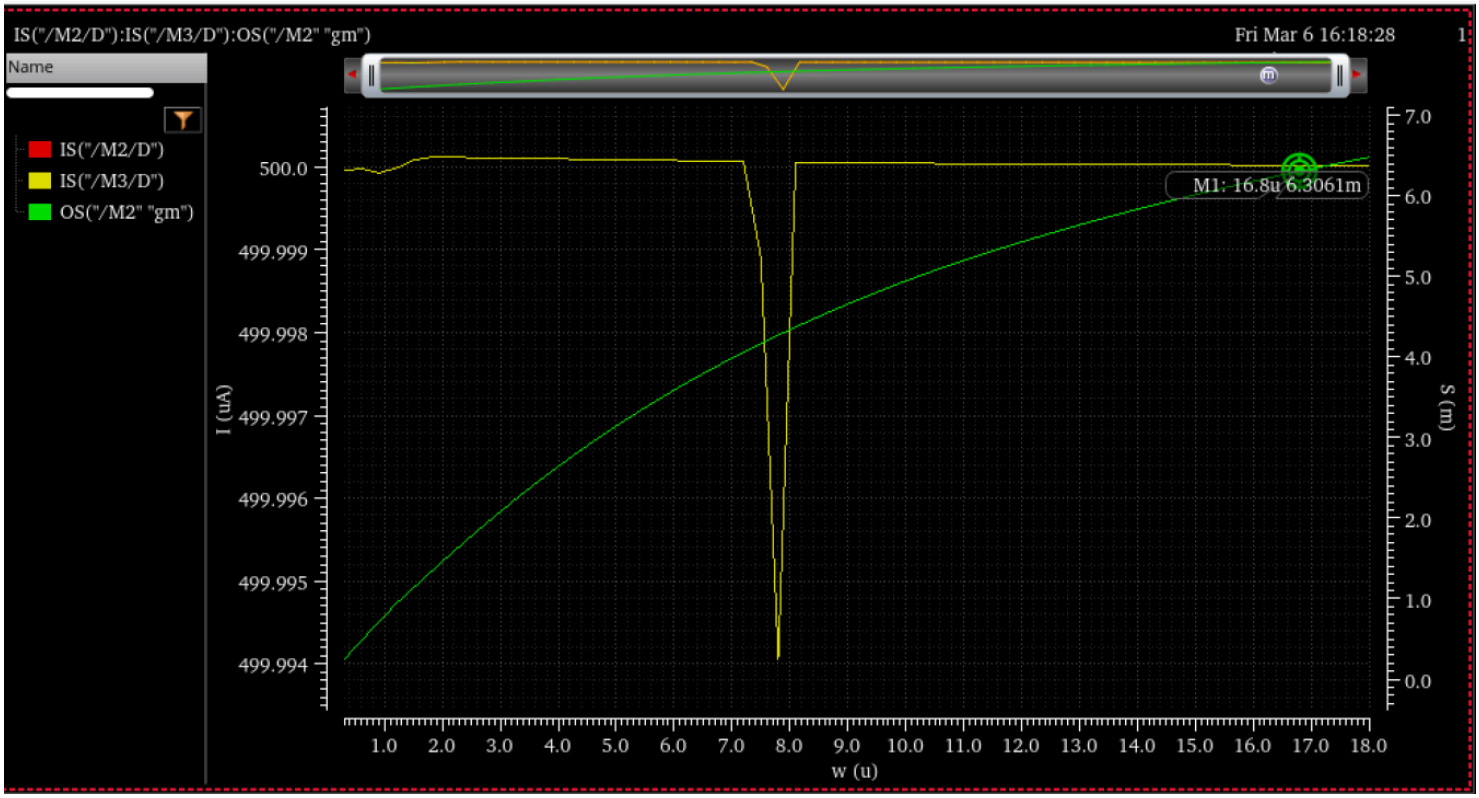
Solve for R

1GHz= 1/1000pi R

1000piR(1G)=1

R=1 / 1000(1G)pi = 318.3ohms

Current & Operating Sweep:



Picking width = 16.8u for Av=2

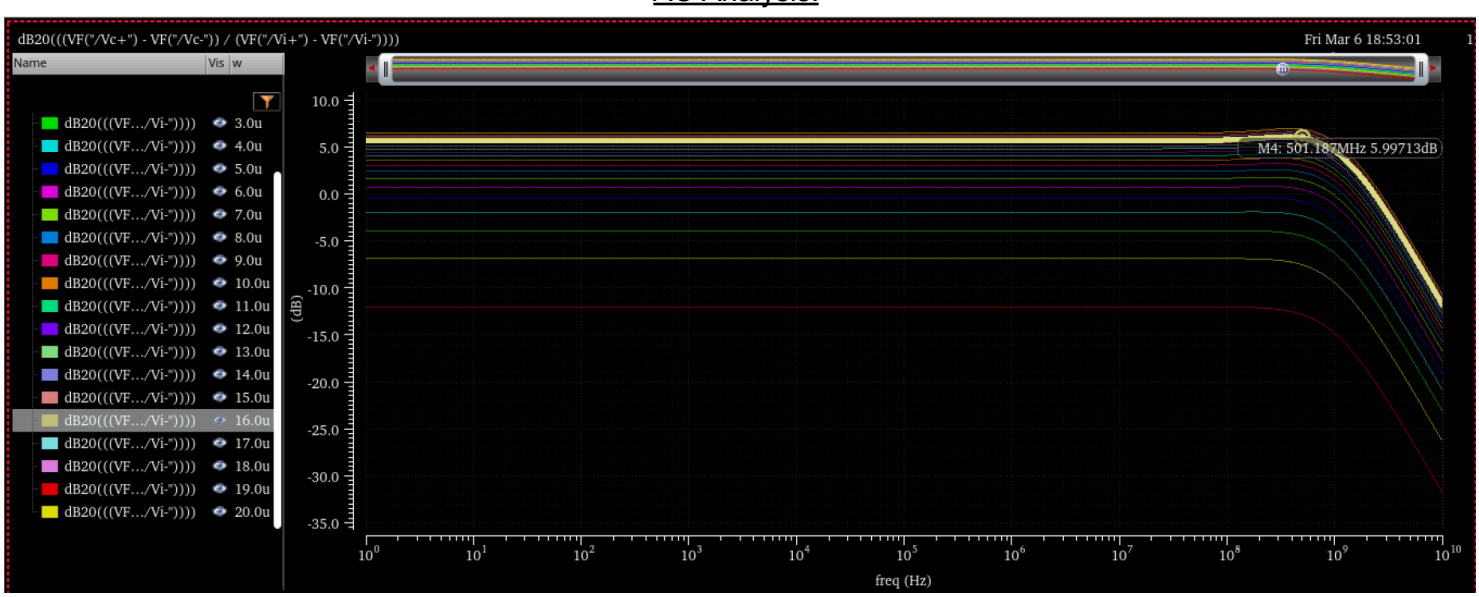
20log(2)=6dB

Av=gm* RD

2=gm * 318.3

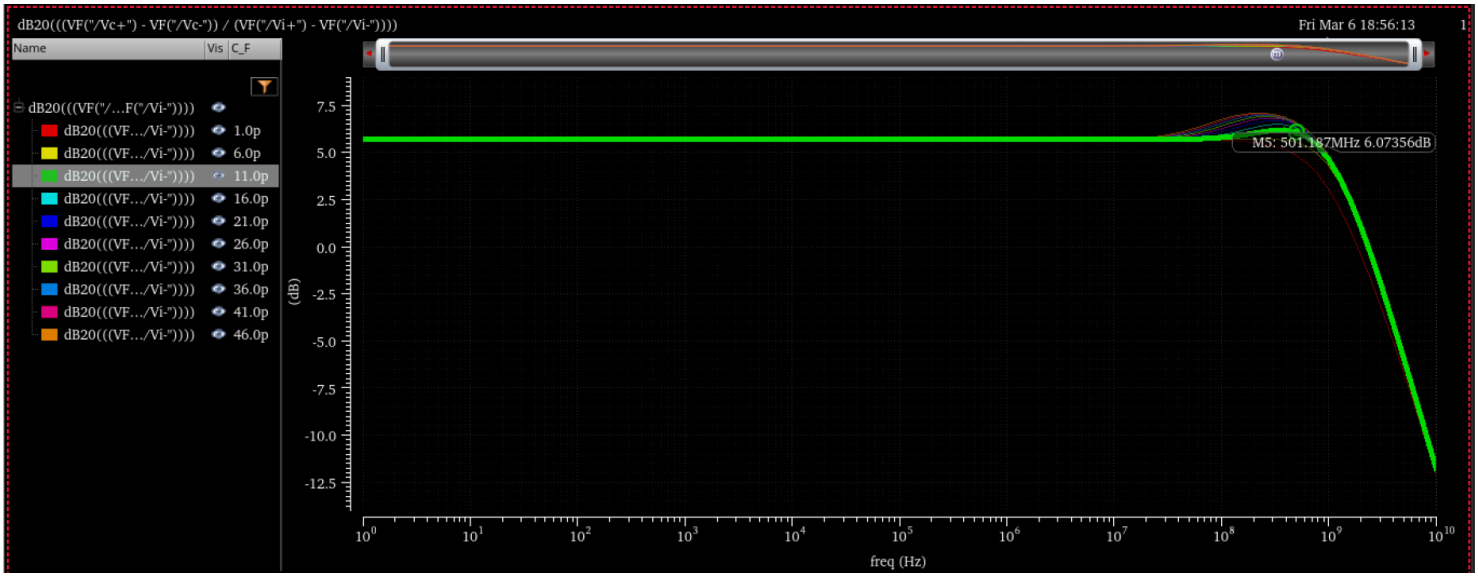
Gm~6.28sm

AC Analysis:



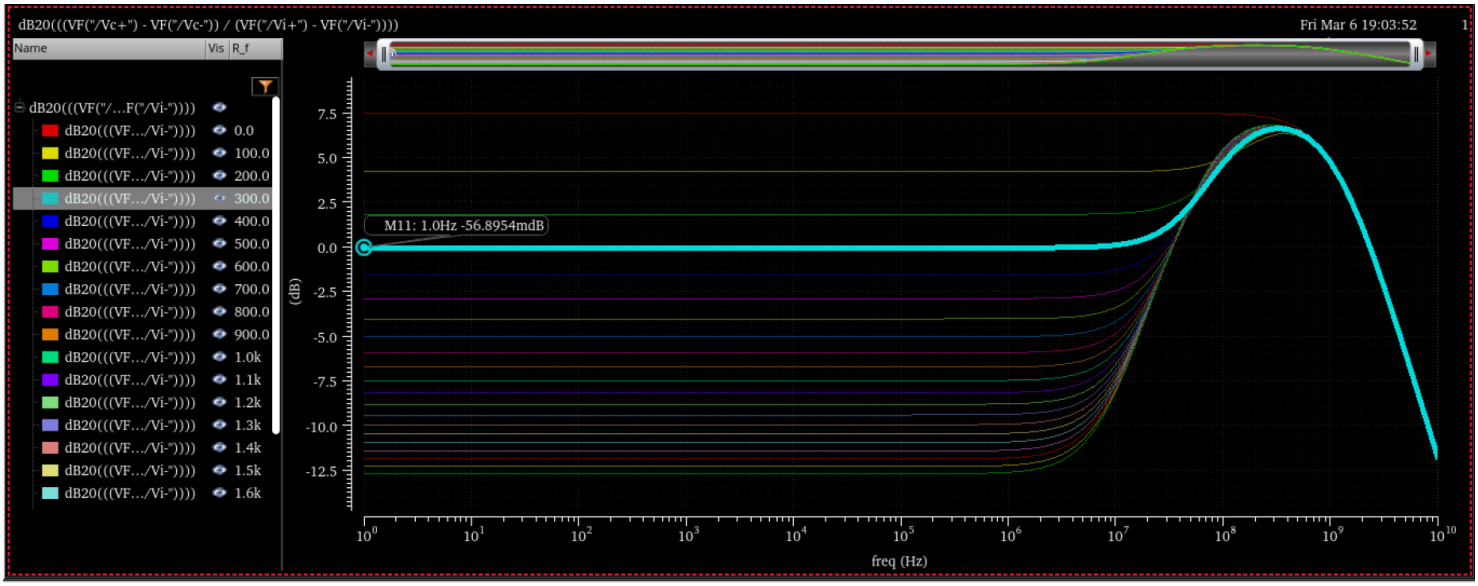
At width 16u we almost get 6dB for 500Mhz, confirming we are pretty close to our calculation!

Capacitor Sweep:



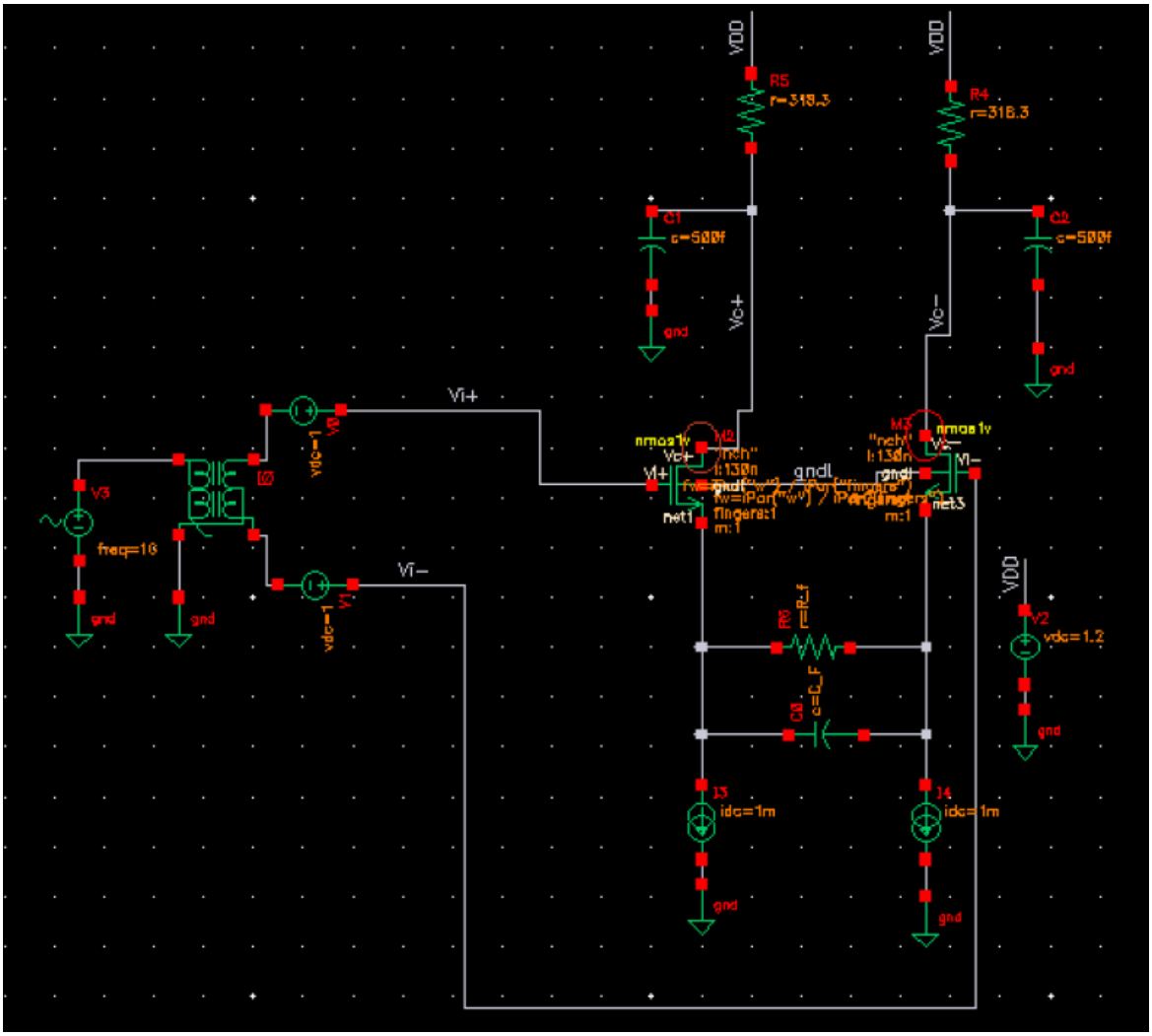
So we get a Cf of ~11pF to attain gain of 6dB at 500Mhz

Resistor Sweep:

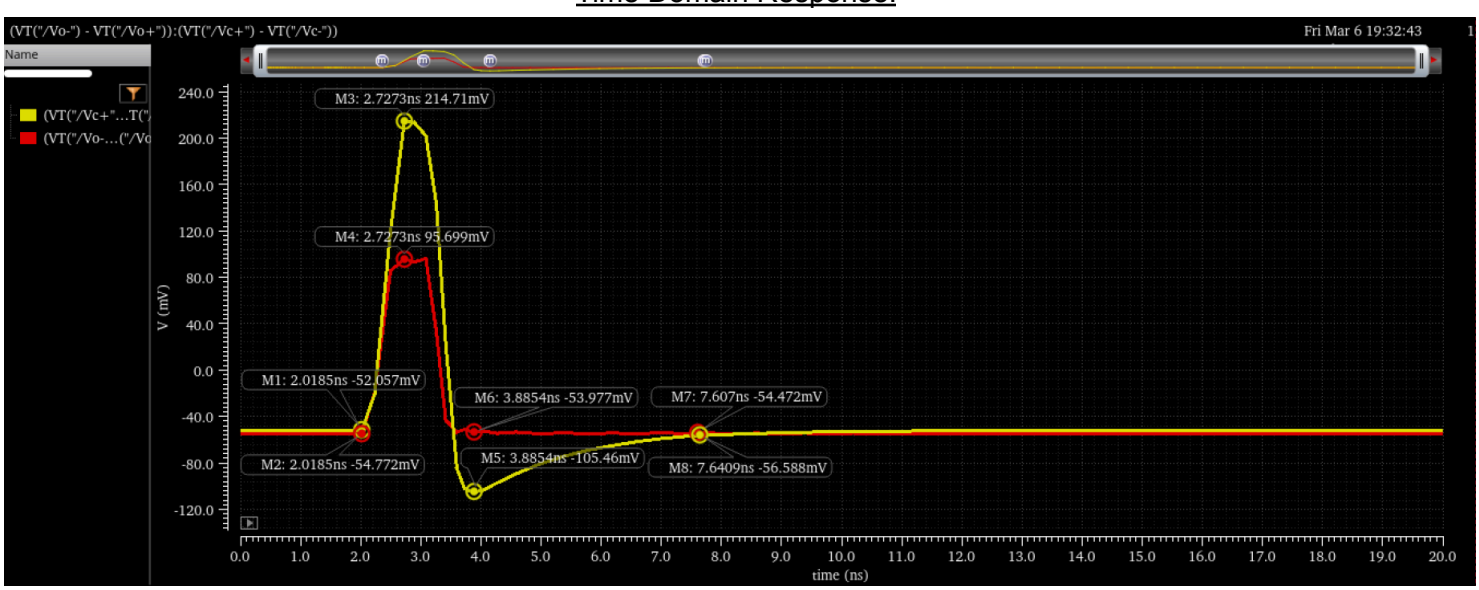


So we get Rf=300ohms for 0 gain at almost DC.

AC Analysis Testbench Circuit:

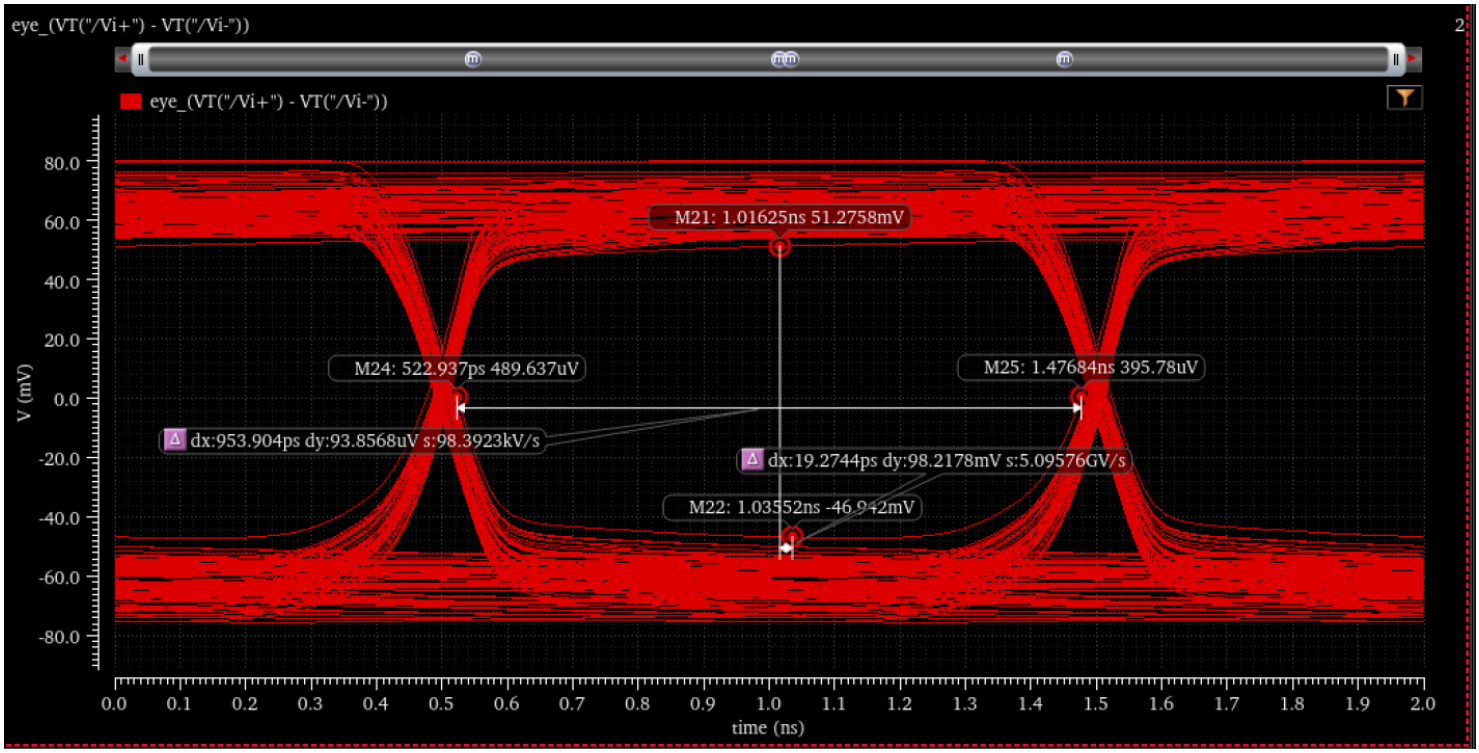


Time Domain Response:



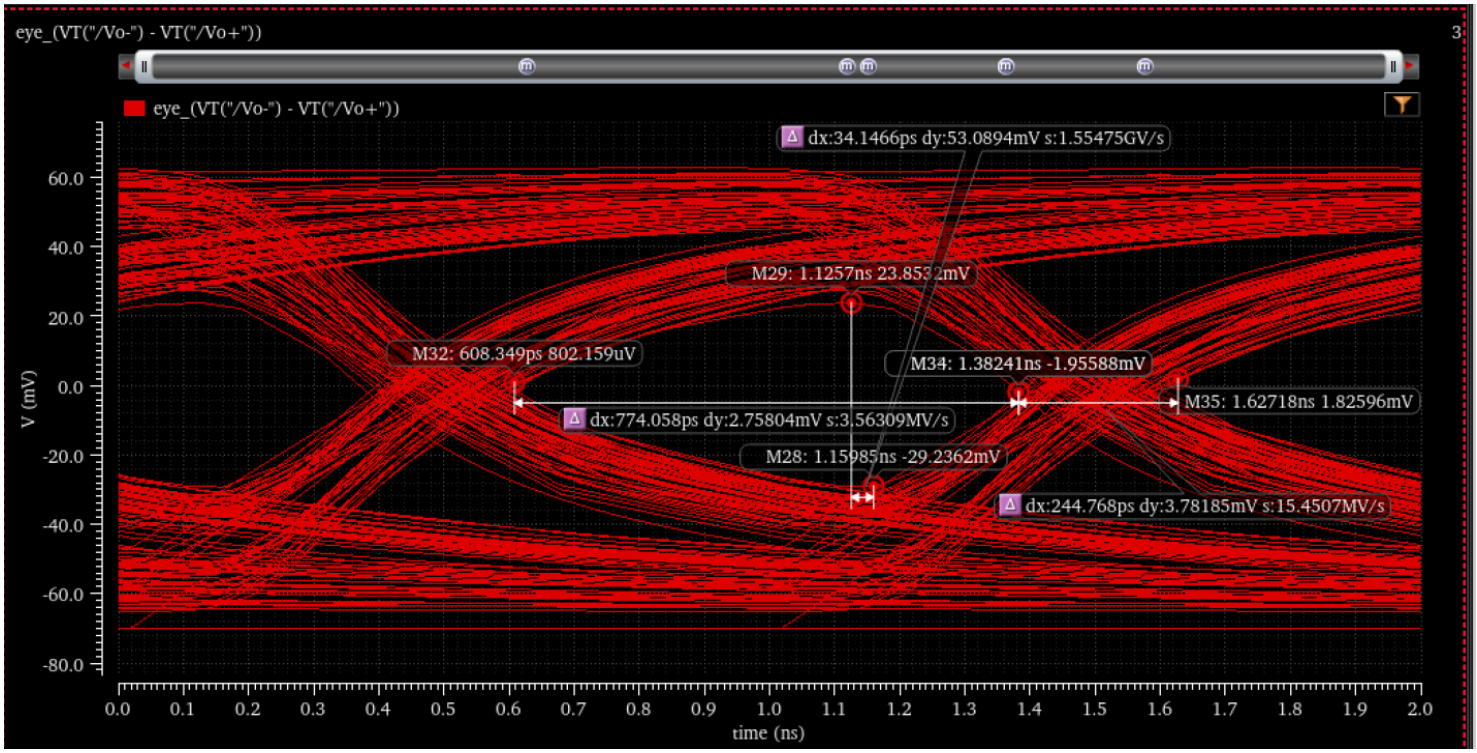
So compared to the original output, the CTLE has double the gain as the Lab 1 output

Input Eye Diagram:



Input eye diagram with 98mV eye opening, period of ~1ns and jitter of 46ps

Output Eye Diagram:



Output Eye Diagram, after channel we have a lossy attenuation down to 53mV, dropping the period down to 774ps, and a jitter of 245ps

CTLE Eye Diagram:

CTLE Eye diagram boosts the output back to 113mV eye diagram, with a period 762ps and a jitter of 241ps. Showing that we have a gain of ~2

Conclusion:

To conclude, I got pretty good results and the lab was successful, as the implementation of the CTLE successfully boosted the output back to a ~113mV eye opening, confirming that the CTLE has double the gain compared to the Lab 1 output. This design successfully corrected the lossy attenuation from the channel, which had dropped the signal to 53mV, and allowed the final output to surpass the original 98mV input eye opening. My simulation sweeps confirmed that the calculated parameters were accurate, specifically showing that a width of 16μm and a Cf of 11pF were sufficient to attain the target 6 dB gain at 500MHz. However, I definitely struggled with transient spikes when switching states in my unit delay cells and significant noise from my delay stages, which resulted in a jitter of 241ps and suggests the pre-emphasis tap sizing wasn't fully optimized.